Quantum × AI for Healthcare Operations

Maximizing Use of Hospital Operating Theatres through Advanced Prediction and Scheduling

Vision & Mission

- **Build the decision layer for healthcare systems**
  Start with OT ⟶ expand to hospital-wide optimization
- **Expand towards other industries that require scheduling**
- **Precision medicine and drug discovery**

We save costs, enhance efficiency, and increase wellbeing —without sacrificing responsiveness to emergencies.

Addressing Critical Pain Points I

Capacity is not the hospital bottleneck ...

... it's the ability to make decisions under uncertainty.

Conservative buffering is a tradeoff for safety against emergency cases and surgery overrun.

30% OT remain idle; Only 50-60% active OT is effectively utilized

Surgical durations are stochastic

Emergency cases are unpredictable

Static Scheduling + Manual Coordination

Surgical resources are being wasted

Addressing Critical Pain Points II

An impossibility for classical algorithms to solve at scale.

Increase Revenue

Currently, 30% OT remain idle, utilization is 50-60% for used OTs

⇩

Increase revenue $30,000/OT/day, $5-20M/hospital/year

Decrease Patient Waiting Time

Currently 2-3 months wait time per patient

⇩

Reduce patient wait time by 20%

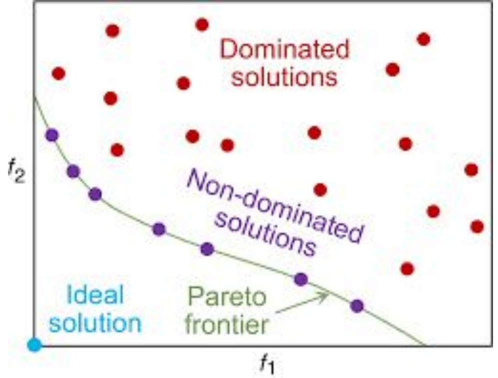
Decrease Staff Overtime

Currently 2-3 hours per day

⇩

Reduce staff overtime by 30%

Beyond Classical Systems:
Quantum-Augmented AI-Driven Multi-Objective Optimization under Uncertainty



Our Approach

Addressing a Multi-Objective Multi-Period NP-Hard Problem under Uncertainty

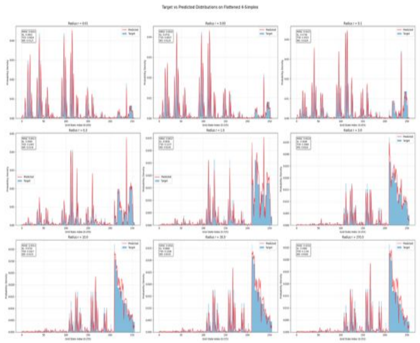
1. **Forecasting**
   quantum kernels ⟶ enhanced accuracy
2. **Multi-objective Optimization**
   quantum computing ⟶ Pareto frontier solutions
3. **Extreme Scenario Modelling**
   ⟶ rare events, emergencies

Quantum × AI Forecasting

Advanced Optimization

Hybrid Architectures
GPU + QPU

Quantum circuit configuration: $n = 8$, $N = 6$, $L = 14$, $T = 2$



At each of these 16 boundary points, we generate 10,000 samples to construct a training dataset. The experiments are implemented on the **IBM Qiskit Simulator**.

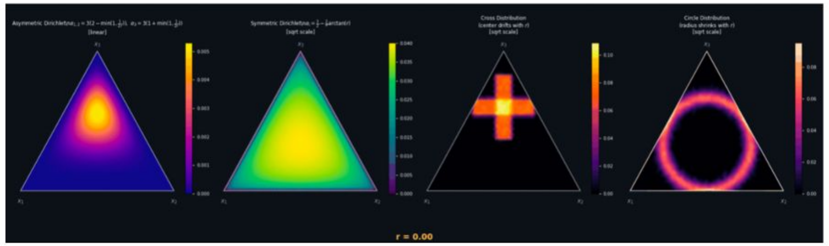
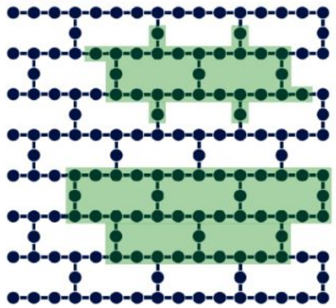


Fig. 1: Graph topologies.

From: Quantum approximate multi-objective optimization



The whole graph shows the coupling map of the 156-qubit ibm_fez device. The green-shaded areas show the 27-qubit graph of the training problem (top) and the 42-qubit graph of the considered MO-MAXCUT problem (bottom) embedded in the ibm_fez coupling map.

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Quantum approximate multi-objective optimization

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System Wide Coordination (OTs, staff & constraints)

More Accurate Forecasting

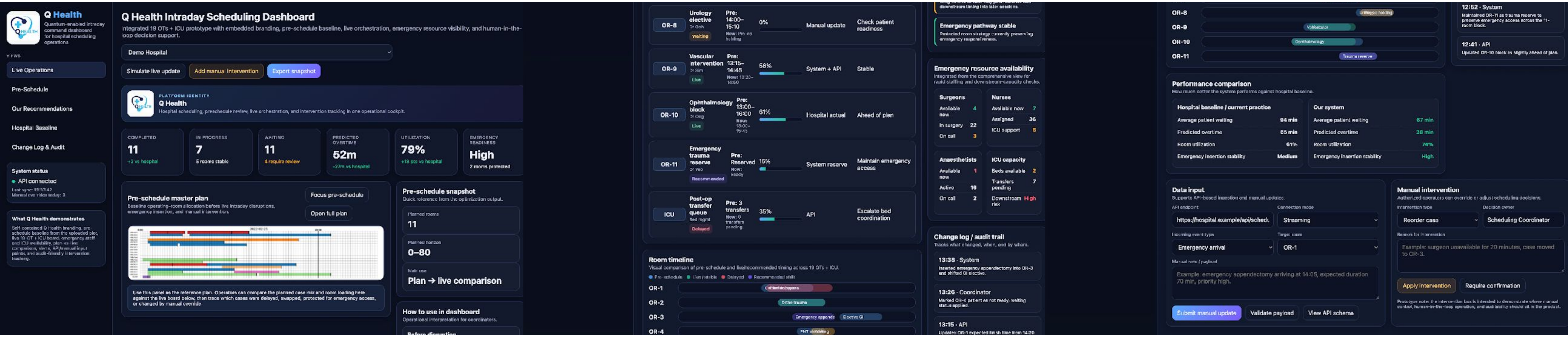
surgery duration & emergencies

Re-Optimization

real time, intra-day

Quantum-Augmented AI Real Time Decision Engine

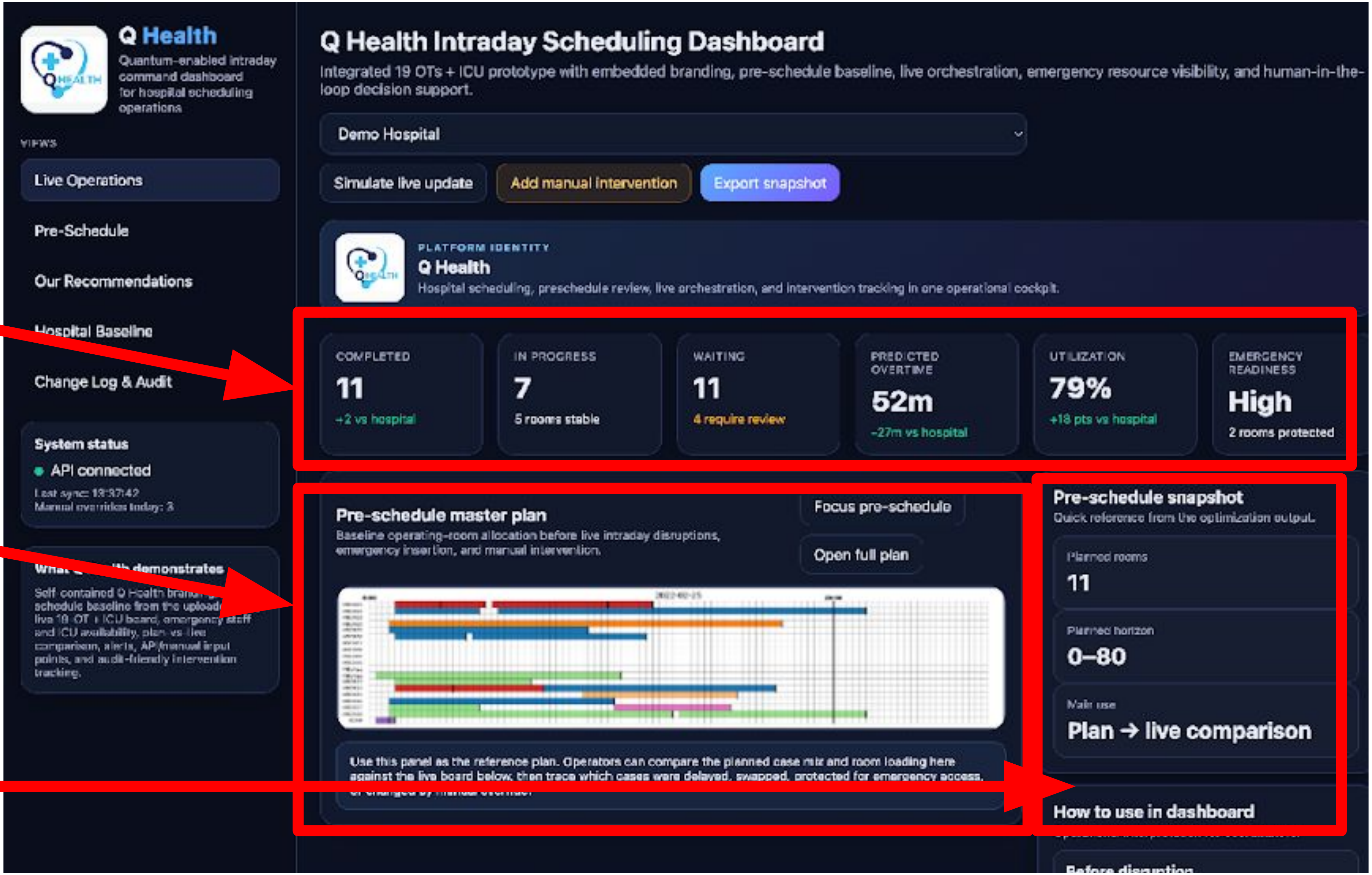
Static Schedules → Real-Time Orchestration under Uncertainty



KPI to measure performance

Baseline Operation Theatre Allocation

Quick Reference from Optimization Output

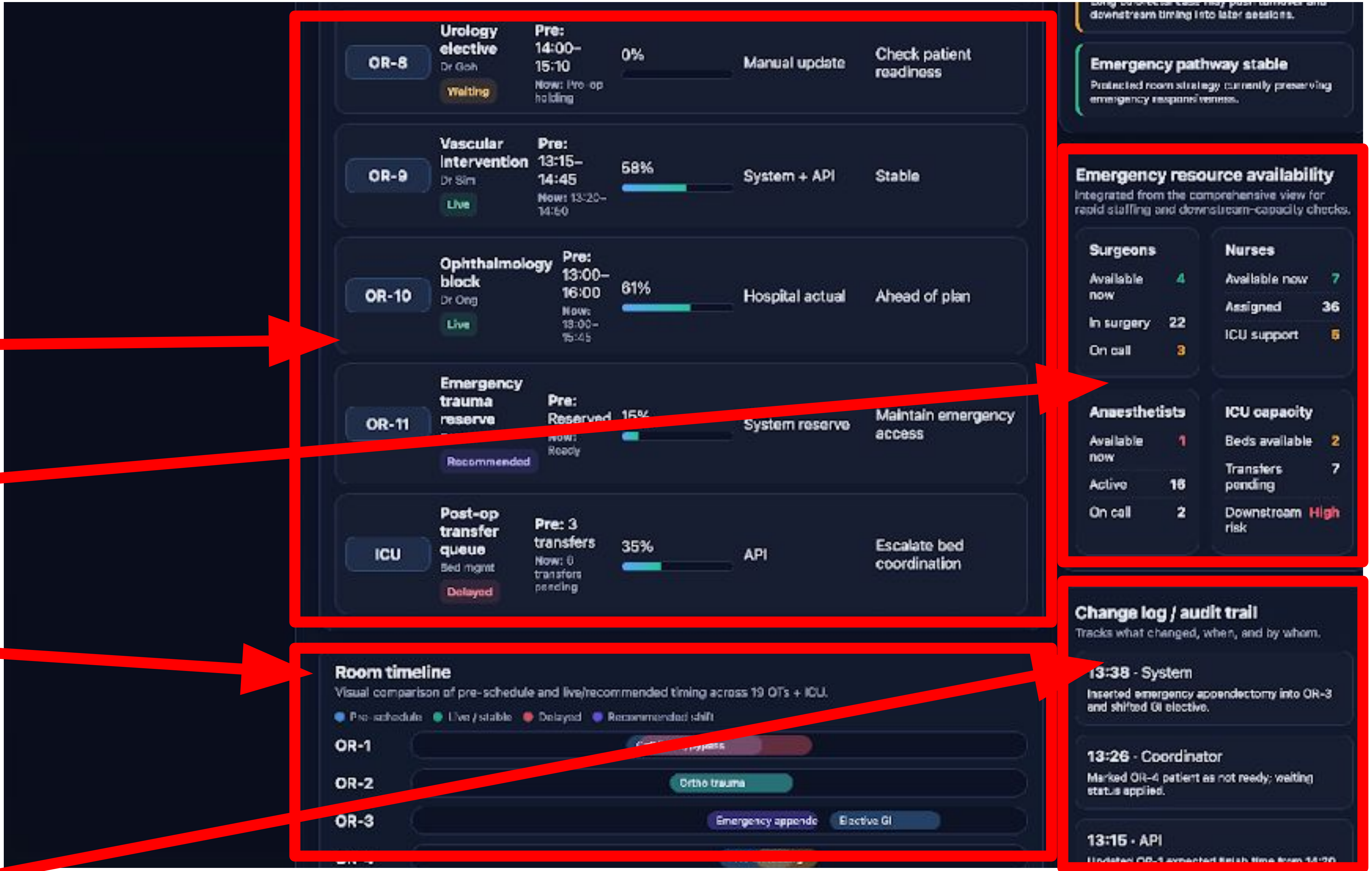


Schedule of Each OT

Capacity Check

Visual Comparison of Pre-Schedule

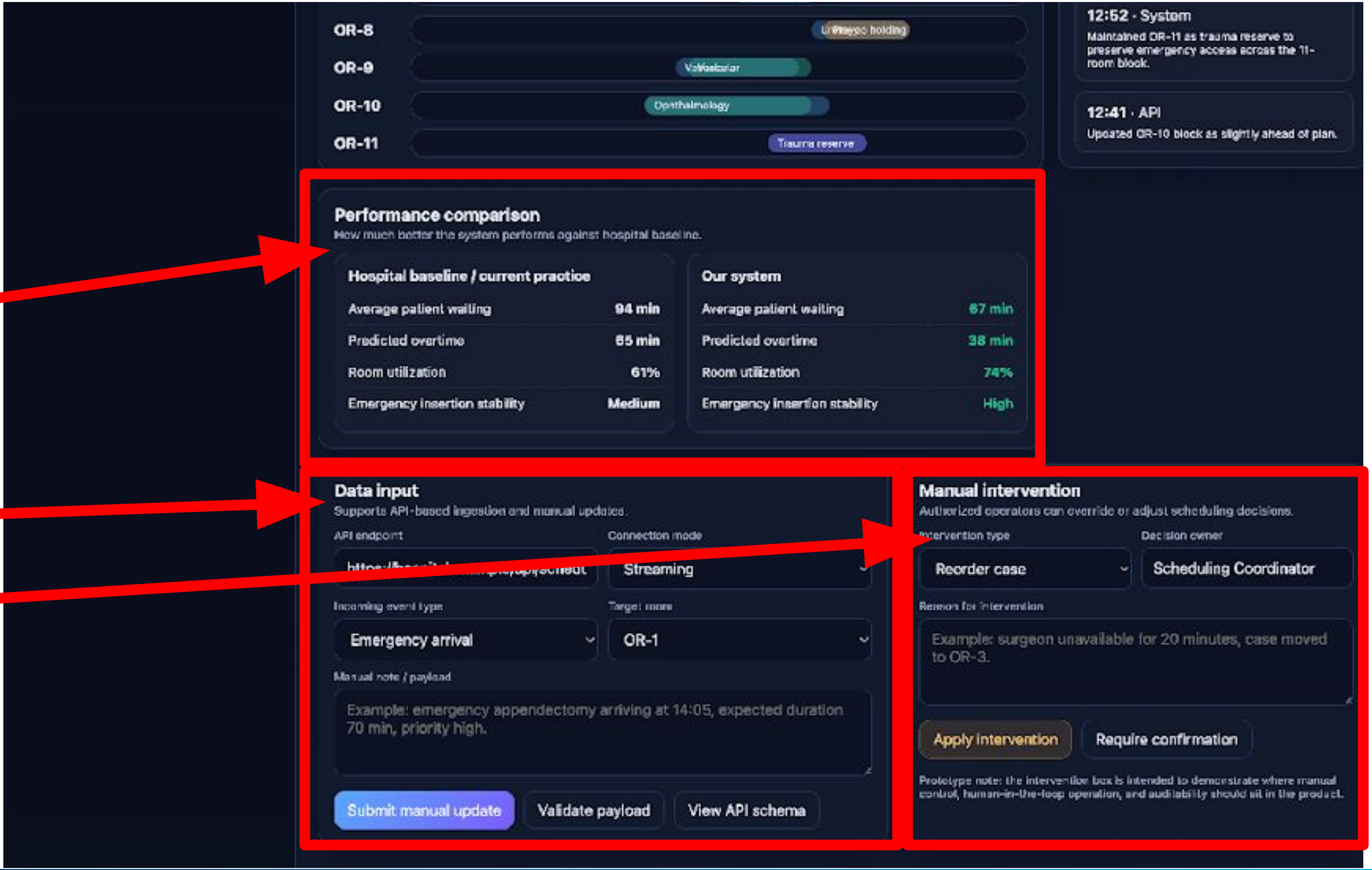
Tracks Changes



How Much Better Q-Health Perform

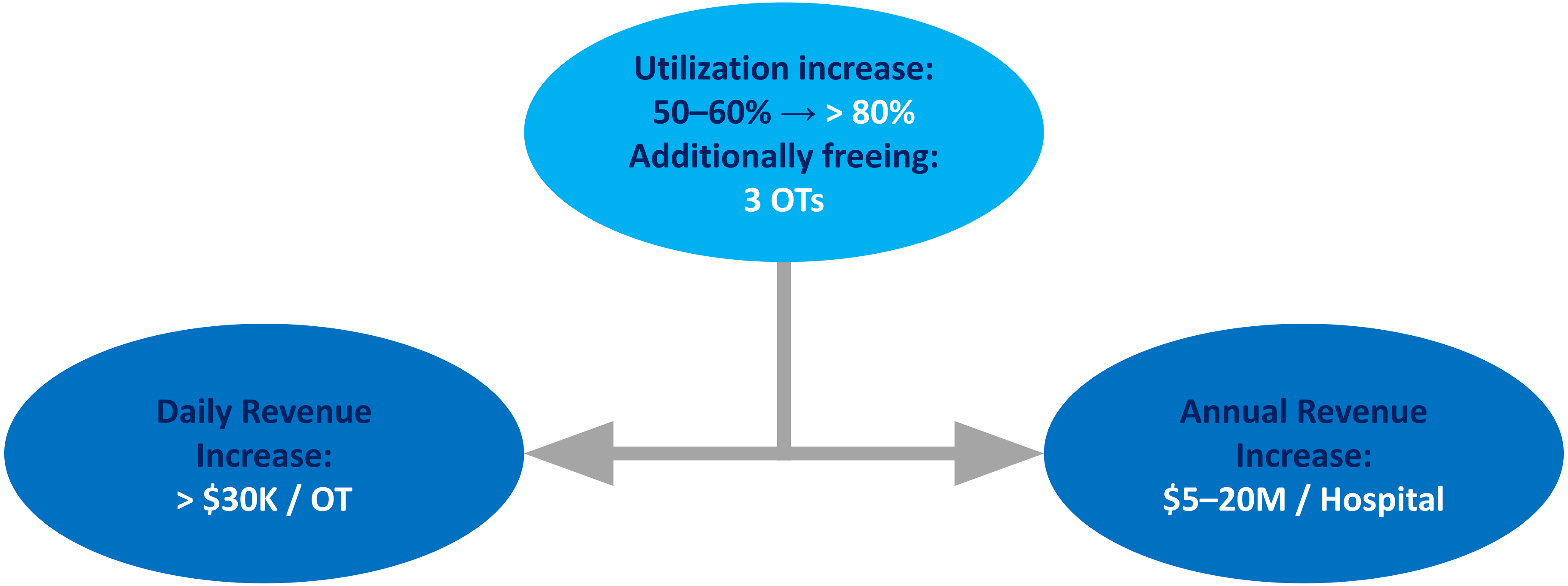
Data Input

Manual Intervention



Validated Impact

Real Hospital Data (19 OTs over Two Years)



Equivalent to Adding Operating Rooms Without Building New Ones

Summary of OT Outcomes & Cost-Benefit Analysis

Target Outcomes within 1 Year with 90% Confidence

- **Increase 30% Throughput**
  Moving from the current 50-60% utilization baseline to >80%, effectively adding the capacity of 3 new OTs without physical construction.
- **Decrease 20% Patient Wait Time**
  Addressing the current bottleneck where patients wait 6-8 months.
- **Decrease 30% Staff Overtime**
  Reducing the current burden of 2-3 hours of daily overtime through better intra-day re-optimization.

Cost- Benefit Analysis

- **Performance-Based Profit Sharing**
  Q-Health shares 50% of the extra profit created
- **Cost**
  $100,000 setup cost + $15,000/OT
- **Benefit**
  $ 5-20 Million / Hospital annually

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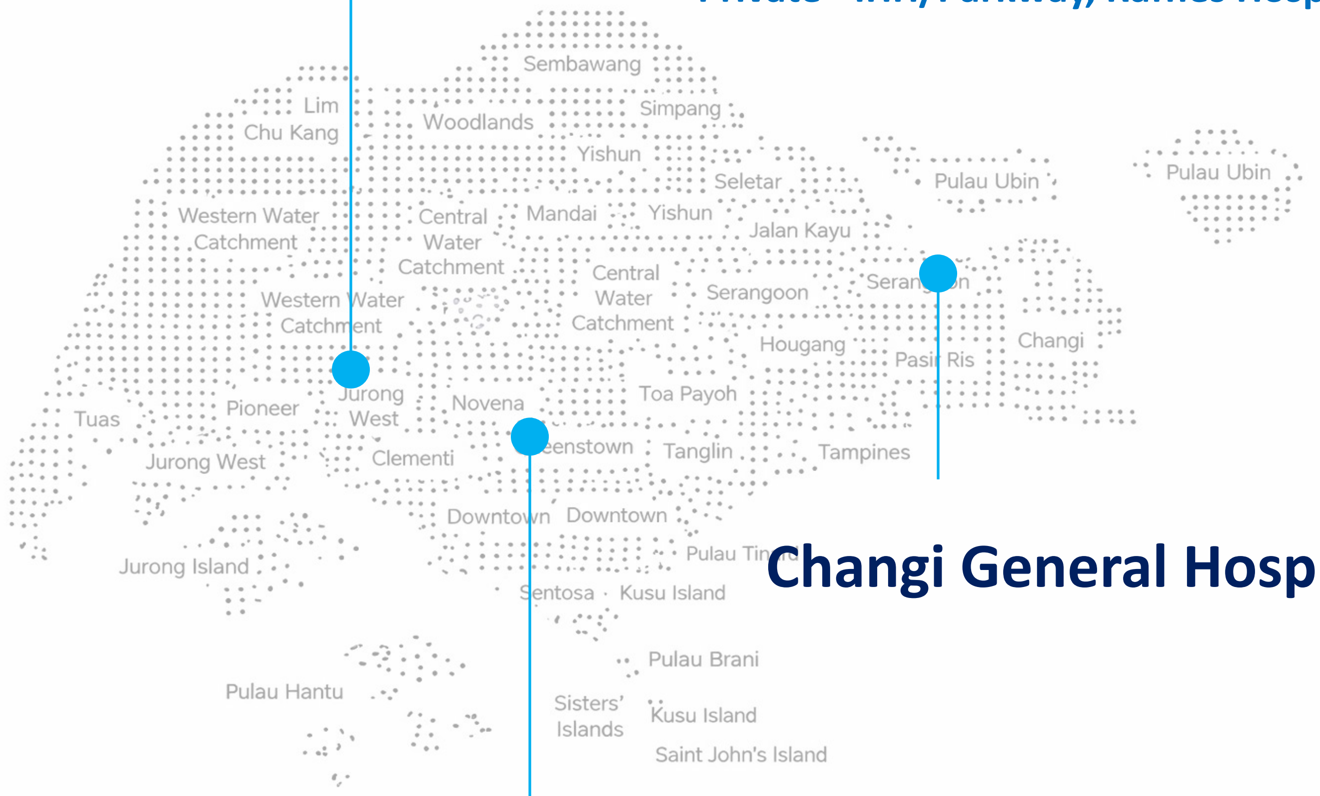
Target Customers

National University Hospital

Public - Singapore 3 healthcare clusters
Private - IHH/Parkway, Raffles Hospital

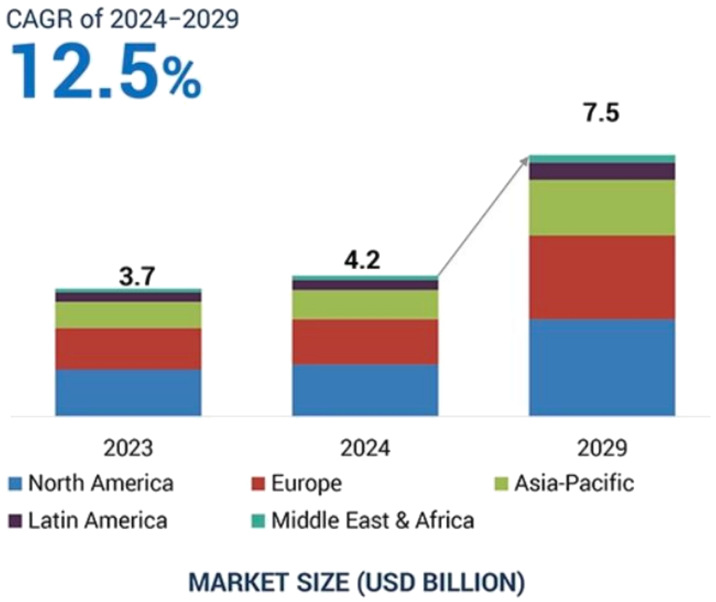


Changi General Hospital

Singapore General Hospital

Provider	Market Role	Scheduling Logic	AI	Quantum Computing	Pricing Model
Q-Health	Quantum-augmented real-time prediction and scheduling engine	Multi-Objective Optimization under Multiple Constraints and Uncertainty	Predict Emergency Occurrence and Surgery Duration. Optimize OT Schedule	Quantum Readiness	$ 90–200K + $ 13–18K/OR/year
Epic (OpTime)	Enterprise EHR Leader	Single-Objective/ Rule-Based	Predict Underutilized Block	None	$500K-1.5M
Oracle Health	Enterprise EHR Leader	Single-Objective/ Rule-Based	None	None	$150-300K
TAGNOS	AI/IOT Real-Time OR Scheduling	Single-Objective/ Rule-Based	Predict Case Length	None	$25-100K 5-15K/OR/Year
GE Healthcare	AI Real-Time OR Scheduling	Single-Objective/ Rule-Based	Advanced Scheduling, OR Capacity Planning	None	$75-200K
Getinge (Torin)	AI Real-Time OR Scheduling	Single-Objective/ Rule-Based	Improve Schedule Reliability	None	$30-150K

Financial Projection and KPIs

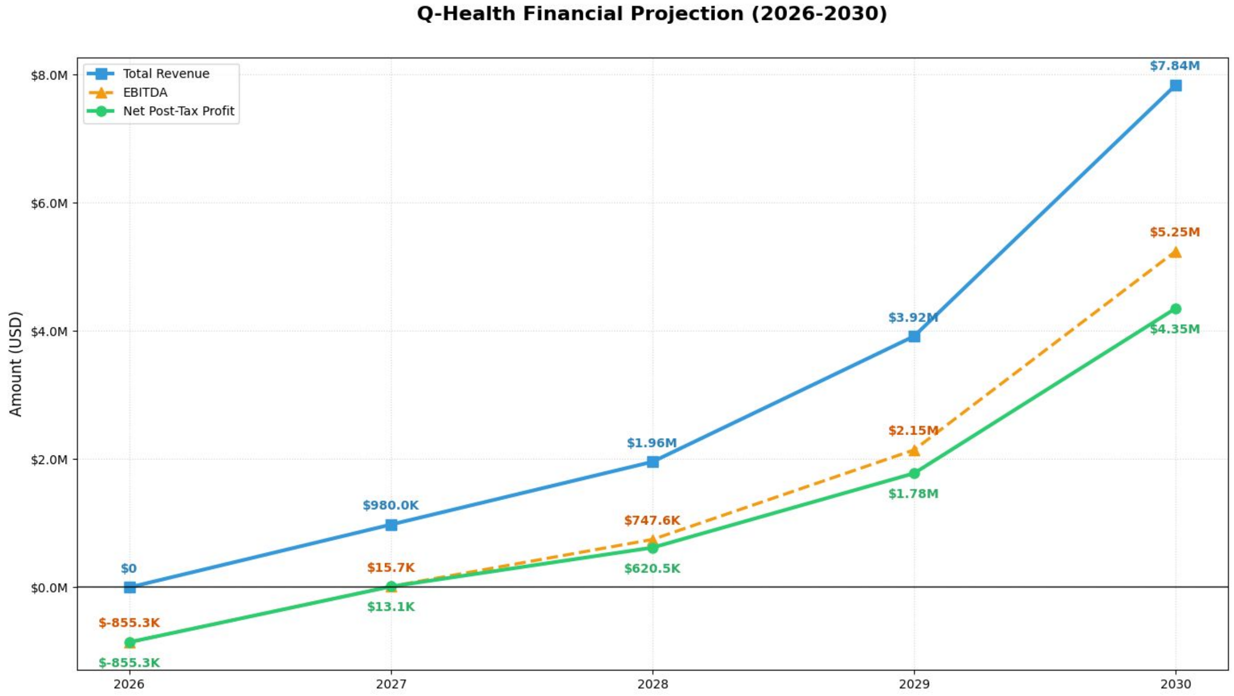


Growth Rate: 100%/year for 4 years

Break Even Point: 2027

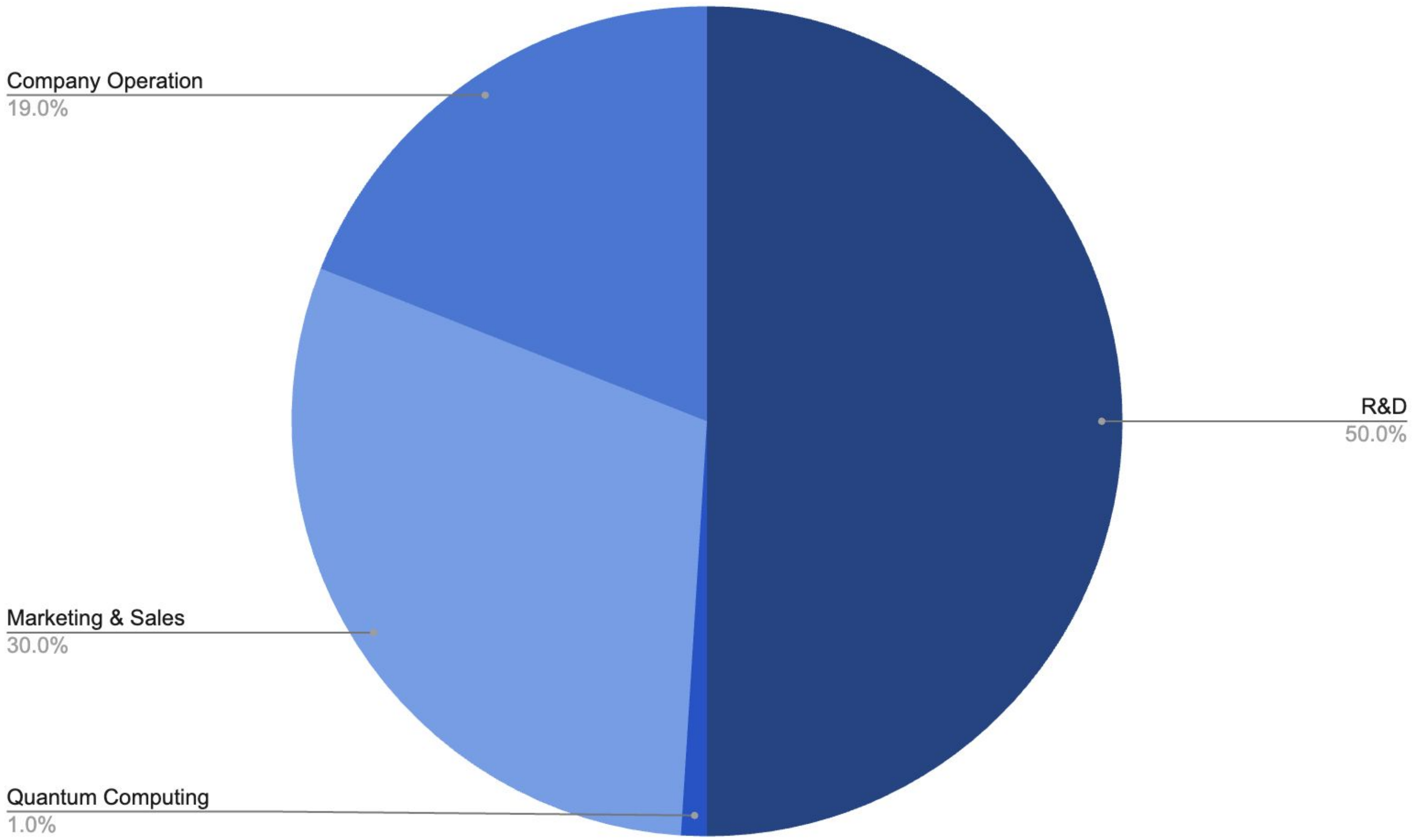
Market Size: $7.5B

CAGR: 12.5%



Asking for $2M in Seed Round

Investment Allocation



Thank You

Any Questions?

QHEALTH